

Mental Health AI Chatbot

Abstract

This project aims to develop an AI-powered chatbot that provides mental health support. It integrates text-based conversation, emotion detection, and meditation timer functionalities, helping users manage stress and improve well-being.

Introduction

Mental health is a growing concern worldwide. Traditional therapy may not always be accessible due to stigma or cost. AI chatbots can provide 24/7 support, offering non-judgmental assistance.

Problem Statement

Many people avoid seeking help for mental health issues. There is a need for an accessible, real-time system that provides support, guidance, and emotional analysis.

Objectives

- Provide instant conversational support
- Offer meditation timer and motivational resources
- Detect user emotions through text
- Ensure privacy and accessibility

System Architecture

The system is built using HTML, CSS, and JavaScript for the frontend, Python (Flask) for the backend, and NLP models for chatbot logic. User messages are processed by Flask, analyzed for sentiment, and appropriate responses are sent back.

Implementation

The chatbot interface allows users to chat in real-time. The Flask backend handles requests and responses. A sentiment analysis model detects emotions, and js-based timer is used for meditation.

Features

- AI-driven chatbot for conversations
- Emotion Detection using NLP
- Meditation Timer for relaxation
- Simple and responsive UI

Results & Screenshots

The chatbot successfully engages in text-based conversations. Emotion detection identifies user sentiment, and the meditation timer supports relaxation. Screenshots of the interface can be

included here.

Advantages

- Available 24/7
- Non-judgmental and private
- Scalable for future improvements

Future Scope

- Add voice-based chatbot support
- Mobile application integration
- Collaboration with psychologists
- More advanced NLP models for empathy

Conclusion

The AI Mental Health Chatbot provides a step toward making mental health support more accessible. With future improvements, it has the potential to become a reliable companion for emotional well-being.